

- 7 $(\text{CH}_3)_2\text{S} \cdot \text{BCl}_3$ is a solid that is commonly used in laboratories as a convenient source of BCl_3 . When heated, it reversibly decomposes to $(\text{CH}_3)_2\text{S}$ and BCl_3 .

Which statement is true?

- A The dative bond is formed using the 2p orbitals of boron and sulfur.
- B $(\text{CH}_3)_2\text{S}$ and BCl_3 act as the Lewis acid and Lewis base respectively in the formation of $(\text{CH}_3)_2\text{S} \cdot \text{BCl}_3$.
- C The dative bond is from boron to sulfur.
- D The C-S-C bond angle decreases when the solid decomposes.